

Question

Execute the command `cat names.txt | grep "Alice"` using an anonymous pipe and understand how data flows between the cat and grep processes

```
GNU nano 2.9.3 names.txt
Alice
John
David
Alice
Robert
Mary

Child process: 5
teja@LAPTOP-FGDPB110:~$ vmstat 1 5
procs -----memory----- --swap-- ----io---- -system-- -----cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo    in   cs  us  sy  id  wa  st  gu
 1  0     0 7356360   5212 196640    0    0   800   17  261    0  0  0 100  0  0  0
 0  0     0 7356360   5212 196640    0    0    0    0  176   68  0  0 100  0  0  0
 0  0     0 7356360   5212 196712    0    0    0    0  159   64  0  0 100  0  0  0
 0  0     0 7356360   5212 196712    0    0    0    0   73   36  0  0 100  0  0  0
 0  0     0 7356360   5212 196712    0    0    0    0   77   36  0  0 100  0  0  0
teja@LAPTOP-FGDPB110:~$ chrt -p $$
pid 329's current scheduling policy: SCHED_OTHER
pid 329's current scheduling priority: 0
teja@LAPTOP-FGDPB110:~$ nano names.txt
teja@LAPTOP-FGDPB110:~$ cat names.txt
Alice
John
David
Alice
Robert
Mary
teja@LAPTOP-FGDPB110:~$ cat names.txt | grep "Alice"
Alice
Alice
teja@LAPTOP-FGDPB110:~$ |
```

Short Theory

An **anonymous pipe** is used for communication between processes. In this command, `cat` reads the contents of `names.txt` and sends its output through the pipe `|`. The `grep` command receives this output and displays only the lines containing **"Alice"**.